HONEYWELL SOFTWARE ENGINEERING TEAMS — IA · BA · PA&T

AI Champions Programme — GitHub Copilot Enterprise, SDD, MCP & Agentic Software Engineering

Module 12

Honeywell Capstone & Adoption Roadmap

Bring the whole journey together: one real use case taken end to end, with evidence on quality, cost and risk — and a 30/60/90-day plan to carry it into your teams.

DURATION
2 hours

FORMAT
Capstone working session

PRIMARY TOOL
Capstone canvas + rubric

How We'll Spend These Two Hours

Recap, brief, build, present, plan. The capstone is where the evidence comes together.

01

Journey Recap & Best Practices

20 min

What you built across eleven modules, and the habits that run through all of it.

02

Capstone Brief & Canvas

15 min

What to deliver, how to choose a use case, and the one-page canvas.

03

Lab: Build Your Capstone

45 min

Integrate your earlier work into one use case with evidence, ROI and governance.

04

Present & Evaluate

25 min

A five-part storyline, a business-case and governance review, and the rubric.

05

The 30/60/90-Day Roadmap

10 min

Turn the capstone into a measurable adoption plan.

06

Next Steps & Wrap-up

5 min

How adoption will be measured and what support follows.

Module 12: The Last Step of the 12-Module Journey

Every earlier module produced something. Today you assemble it.

FOUNDATION			BUILD & VALIDATE			EXTEND WITH TOOLS & AGENTS				GOVERN	CAPSTONE
1	2	3	4	5	6	7	8	9	10	11	12
Objectives & GenAI Recap	Copilot Advanced	Prompt & Context Eng.	Spec-Driven Dev (Spec Kit)	SDD to Full SDLC	Test Strategy & Validation	MCP & Custom MCP Server	PR Automation & LLM-Judge	No-Code Agents (Rovo)	Code Agents & Simple RAG	Observability, ROI & Gov.	**Capstone & Roadmap**

M01–03

Copilot power use and context practices behind every later step

M04–06

Specs, SDLC paths and a test strategy that give you evidence

M07–10

MCP workflows, PR gates and agents that extend the work

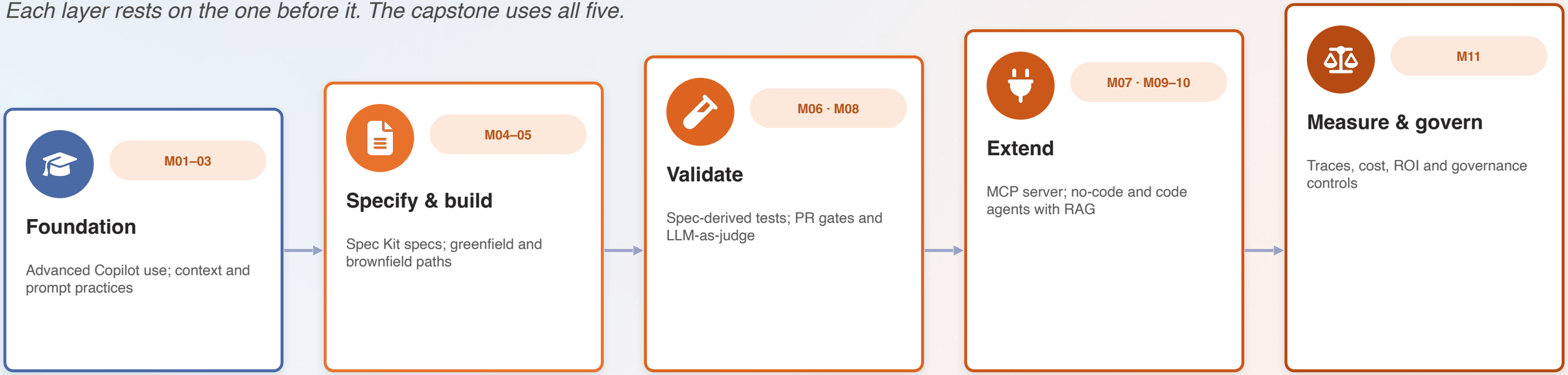
M11

Observability, ROI and governance that make it credible

The capstone principle: integrate, don't restart. Link the artefacts you already made; spend the time on the gaps and on the business case.

What You Now Have: Five Layers of AI-Assisted Engineering

Each layer rests on the one before it. The capstone uses all five.



Look back before you look ahead: you have a spec, a test strategy, quality gates, tools, agents and a way to measure them. Few teams have all of that in one place.

Eight Habits That Run Through the Programme

The tools will change. These habits will not.

M04–05

Specs before code

Agree acceptance criteria first; trace work back to them

M06

Tests from the spec

Derive tests from criteria; see them fail first

M07

Least privilege

Give agents and tools only what the job needs

M08

Evidence in review

Comments with location, evidence and confidence

M09

Drafts before autonomy

Start agents at suggest or draft; a person decides

M10

Limits and logs

Step, cost and permission limits; log every tool call

M11

Count accepted work

Measure cost per accepted output, not output volume

All

Name an owner

Every workflow has an owner and a review date

Use it as a checklist: in your capstone review, ask which habit each artefact demonstrates — and which habit is missing.

Best Practices for Taking This Into Your Teams

Start small, prove it, then widen.

DO THIS	AVOID THIS
Start with one bounded use case and a measured baseline	Roll out every tool to every team at once
Keep specs, tests and prompts in the repository, versioned	Treat AI output as accepted because it looks right
Keep a human approving every merge and risky action	Grant broad access or auto-approve write actions early
Reuse approved MCP servers and agents before building new ones	Report speed gains without rework and cost
Record owner, permissions and an off switch for each workflow	Leave workflows with no owner or review date
Review scale, tune, restrict or retire on a schedule	Wait for perfection before piloting anything

Golden rule: adoption is a series of small, measured decisions — not a launch event.

The Capstone: One Use Case, End to End

Eleven pieces in four groups. Most already exist from earlier labs — link them, then fill the gaps.

Define

- Use case and users
- Solution blueprint
- Specification and plan

Build & prove

- Implementation approach
- Test strategy and evidence
- PR and judge evidence

Extend

- MCP workflow or server
- Agent approach: no-code and/or code with RAG

Measure & plan

- Monitoring design
- ROI model
- 90-day adoption roadmap

Final review covers: the business case, governance controls and an enterprise rollout roadmap — so evidence and risk matter as much as the demo.

The One-Page Capstone Canvas

If it does not fit on one page, it is not yet clear enough to adopt.

Use case & users

Who benefits, what job

Path

Greenfield or brownfield

Spec & criteria

Acceptance criteria, traced

Test evidence

Strategy, results, gaps

Quality gates

PR checks and judge

Tools & agents

MCP, no-code, code + RAG

Monitoring

Traces, signals, limits

ROI model

Baseline, cost, assumptions

Risks & governance

Owner, access, approvals

30/60/90 roadmap

Actions and checkpoints

Fill it in this order: top-left to bottom-right follows the journey. Gaps you find on the way are the capstone's most useful output.

Choosing a Good Capstone Use Case

The best capstone is small enough to finish and real enough to matter.

A GOOD CHOICE IS…

- A real job someone on your team does repeatedly
- Bounded: a feature, a change or one workflow
- Measurable: you can state a baseline today
- Ownable: a named person will carry it on
- Safe to start: reversible, non-production first

AVOID…

- A whole platform or a multi-team programme
- No baseline and no way to measure it
- Needs production write access on day one
- Depends on data you may not use
- Chosen only because it demos well

Greenfield or brownfield: either is fine. A brownfield change with regression protection often shows more of what you learned than a new application does.

Lab: Build Your Capstone

LAB 1 · 45 min

Integrate your earlier work into one use case, with evidence, ROI and governance.

Goal Complete the capstone canvas for one use case, link the evidence you already produced, and draft the ROI model and 30/60/90 roadmap.

STEPS

1. Pick the use case and path using the criteria. Fill the top of the canvas: users, acceptance criteria and the baseline you can state today.
2. Build an evidence index that links, not recreates, your spec, test strategy, PR and judge results, and MCP and agent artefacts.
3. Complete monitoring design, ROI model (assumptions stated) and the governance record: owner, access, approvals, off switch.
4. Draft the 30/60/90 roadmap and rehearse the five-part presentation with your team.

FILES YOU CREATE

```
repo/
lab12/
  capstone-canvas.md
  evidence-index.md
  roi-model.md
  governance.md
  roadmap-30-60-90.md
```

DONE WHEN

A reviewer can follow the canvas from use case to roadmap, every claim links to evidence, and the ROI model lists its assumptions.

How the Capstone Is Evaluated

Seven criteria. Strong work shows evidence, not just a working demo.

CRITERION	STRONG WORK SHOWS
Spec & traceability	Acceptance criteria are clear and every artefact traces back to them
Test evidence	Tests derive from the spec; results and gaps are shown honestly
Quality & PR controls	Gates and review are defined; humans own approval
Tools & agent design	Narrow tools, sensible limits, drafts before autonomy
Governance & risk	Owner, access, approvals and an off switch are documented
Measurement & ROI	Baseline, cost per accepted output and assumptions are stated
Roadmap realism	Small, owned steps with checkpoints and a decision rule

Illustrative rubric: confirm the final criteria and scoring with your facilitators before presenting.

Presenting Your Capstone: A Five-Part Storyline

Lead with the problem and the evidence; keep the demo short.

1. **Problem**
   Who, what job, today's baseline
2. **Approach**
   Path, spec, tools and agents used
3. **Evidence**
   Tests, review results, traces, gaps
4. **Value & risk**
   ROI with assumptions; controls
5. **The ask**
   30/60/90 plan and decisions needed

TIPS FOR THE ROOM

- **Show one failure and what you learned**
- **Say what you did not measure**
- **Make the ask specific and small**

Credibility comes from honesty: leaders trust a plan that names its gaps more than one that claims none.

The Final Review: Business Case and Governance Controls

Two questions every reviewer will ask: is it worth it, and is it under control?

BUSINESS CASE

Baseline	How the work is done today, measured
Cost	Tokens, tools and human review time
Value	Time saved and rework avoided, from evidence
Assumptions	Written down, with what would change them

GOVERNANCE CONTROLS

Ownership	A named owner and review date
Access	Least privilege for tools and agents
Approvals	Humans approve merges and risky actions
Control	Monitoring, limits and an off switch

Link, don't invent: every number in the business case should point to a trace, a record or a stated assumption — reuse the Module 11 worksheet.

The 30/60/90-Day Adoption Roadmap

A template to adapt: each phase has a focus, a few actions and a checkpoint that decides what happens next.

Days 1–30 — Prepare & baseline

- Name owners and approve tools
- Capture the baseline for the use case
- Put instructions, specs and prompts in the repo

Checkpoint: baseline recorded, pilot scope agreed

Days 31–60 — Pilot & prove

- Run the workflow with a small group
- Track accepted output, review time, cost
- Fix leaks and gaps found in traces and review

Checkpoint: evidence reviewed with the owner

Days 61–90 — Decide & extend

- Scale, tune, restrict or retire the workflow
- Share reusable specs, skills and servers
- Plan the next use case from the evidence

Checkpoint: rollout decision and next plan

Adapt it: the dates are a frame, not a rule. Move the checkpoints to match your release cycle and who has capacity.

How Adoption Will Be Reviewed After Training

Six signals from the post-training assessment framework. Capture a baseline now.

Feature implementation cycle time

Time from accepted spec to merged change

PR review time

Time and rounds needed to review a change

Regression & defect signals

Escaped defects and regressions after release

Reusable skill & spec adoption

How many teams reuse the shared assets

Token spend discipline

Usage against limits; leaks found and fixed

Early ROI signals

Cost per accepted output in piloted workflows

Baseline first: you can only show a change against a starting point. Record today's values before the pilot begins; targets come from your own history.

Next Steps and Support After Today

You leave with a plan; the programme includes follow-up to help it stick.

Today

Capstone and 30/60/90 roadmap agreed with your team

30-day catch-up review

What you applied, what is deployed, where adoption is blocked

Masterclass

Advanced lessons, common mistakes, SDD maturity, testing, next patterns

Gap-closure recommendations

Based on progress, leadership expectations and blockers

YOUR FIRST WEEK

- **Record the baseline**
- **Name an owner per workflow**
- **Book the pilot group**

Bring evidence: come to the 30-day review with what you applied, what you measured and where you are stuck — that is what makes the session useful.

What You Can Do Now

Six outcomes — and one plan to put them to work.

- ✓ Take one use case end to end with spec, tests, review and tools
- ✓ Choose and present a capstone with a clear storyline
- ✓ Build a credible business case with stated assumptions
- ✓ Document governance controls: owner, access, approvals, off switch
- ✓ Write a 30/60/90-day roadmap with checkpoints and decisions
- ✓ Name the signals you will use to review adoption

NEXT: YOUR 30-DAY CATCH-UP REVIEW

Bring what you applied, what you measured and where adoption is blocked.

THE JOURNEY IN FIVE WORDS

Specify → Build → Validate → Extend → Govern

Questions & Discussion

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